VERITRUST

Four-Core Detection Architecture Blueprint

HF-managed production revision

ARCHITECTURE	HF-MANAGED	EVIDENCE-FIRST	PRODUCTION GATES

Authoritative decision. VeriTrust executes no learned production inference in browser code, Vercel functions, platform workers, local runtimes, or VeriTrust-owned model servers. Learned production inference uses Hugging Face-managed infrastructure. The canonical hierarchy is HF-operated hf-inference when the exact model/task and all gates pass; then Hugging Face Inference Endpoint; then private Hugging Face repository plus Hugging Face Inference Endpoint. Third-party Hugging Face Inference Providers are DISABLED_BY_DEFAULT, outside the authoritative production path, and never an automatic fallback.

Status: implementation architecture and research package; no production performance claim. All UNKNOWN values remain explicit.

Prepared from the executable repository, seven historical PDFs, fresh official Hugging Face research, primary competitor sources and C2PA standards.

Contents and control

This is an Implementation Architecture Handbook, not a standalone executable specification. Critical invariants, model identities, deployment classes, evidence semantics, receipt fields, lifecycle states and companion references are generated from validated canonical structured sources; all provider availability is point-in-time and promotion-gated.

NORMATIVE IMPLEMENTATION COMPANIONS: Architecture_Source_of_Truth.yaml; Model_Registry_Proposal.json; HuggingFace_Deployment_Matrix.csv; Implementation_Backlog.csv; Database_Target_Contract.md; Verification_Test_Plans.md; Migration_Roadmap.md.

Cross-document invariants

- HF-managed learned inference only; no normal third-party provider tier.
- Registry resolution precedes dispatch.
- Receipt identifies exact deployment, bundle and ordered learned components.
- raw_output, raw_label, raw_score, normalized_label, normalized_score, calibrated_probability and calibration_version remain distinct.
- Expected-evidence lifecycle uses the canonical 16-state enum: planned, dispatched, running, completed, abstained, not_applicable, unavailable, timed_out, failed, cancelled, privacy_blocked, policy_skipped, insufficient, unsupported, degraded, budget_exceeded.
- Specialist evidence cannot silently substitute or own final policy.
- No original PDF or production source file is overwritten.

1. Executive architecture decision

Specialists produce evidence. Only the versioned Gateway policy produces a final trust decision.

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2. Repository-grounded current state

The baseline contains 180 original files; 158 textual files and 36,672 lines were indexed. Node 24 has one runtime dependency (busboy). Seven Vercel functions are capped at 60 seconds. No local ML framework or production model weights are present.

Current remote inference is configuration-driven through mutable owner/repository aliases and a generic HF router/legacy fallback. Direct routes allow model/provider/local fallback; Gateway link/phishing adapters disable some fallback, while deepfake still substitutes Pixel/Prism.

Verified strength	Release blocker
Tenant HMAC, idempotency, signed uploads, SSRF/webhook controls	Mutable aliases and post-execution provenance lookup
Artifact/evidence/policy concepts	Silent model or local heuristic substitution
Current CLI and PowerShell match API	Uncalibrated scores and wrong URL label map

3. Target trust boundaries and four-core architecture

The four cores are Link, Phishing, Deepfake and Learning, coordinated by the Unified Gateway. Link/Phishing/Deepfake emit specialist evidence. Learning is a deterministic domain subsystem, not a verdict engine.

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CLIENT -> VERITRUST INGRESS -> IMMUTABLE ARTIFACT -> DETERMINISTIC FEATURES
-> PERSIST EXACT REGISTRY RESOLUTION -> HF-MANAGED INFERENCE
-> VALIDATE RESPONSE + RECEIPT -> SPECIALIST EVIDENCE
-> SUFFICIENCY + CORRELATION -> VERSIONED POLICY -> DECISION / HUMAN REVIEW
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Boundary	Required controls
Client -> VeriTrust	auth, tenant binding, idempotency, MIME/magic/size/deadline
VeriTrust -> HF	TLS, scoped token, egress allowlist, exact deployment/revision, response cap
HF -> VeriTrust	schema/label/deployment/receipt validation
VeriTrust -> webhook	destination allowlist, DNS rebinding defense, HMAC, replay window

4. HF-managed inference strategy

The hierarchy and complete model-role table below are loaded from the canonical YAML, registry JSON and deployment CSV. Third-party Inference Providers remain disabled by default, outside the authoritative path, and never an automatic fallback.

Order	HF-managed mode	Eligibility
1	HF-operated hf-inference when the exact model/task and all gates pass	Exact model and task have a fresh, explicit hf-inference mapping; golden contract, privacy, latency, quota, and cost gates pass.
2	Hugging Face Inference Endpoint	Default for public models without verified serverless support; commit revision, endpoint deployment, and response schema are pinned.
3	private Hugging Face repository plus Hugging Face Inference Endpoint	For VeriTrust fine-tunes, private learned heads, GBDT/ONNX models, or controlled copies with custom handlers.

Role	Canonical model key / repository / revision	Deployment	Status
face manipulation	deepfake.face_manipulation.yermandy_v1 | yermandy/deepfake-detection @ 9a6857ec642deb57373c5437be803a199468b8c6	private HF repo + HF Inference Endpoint	research_candidate
face detection landmarks alignment	deepfake.face_pipeline.yunet_private_v1 | veritrust-private/deepfake-face-pipeline-yunet-v1 @ UNKNOWN	private HF repo + HF Inference Endpoint custom container	requires_packaging_and_qualification
independent low-level forensics	deepfake.forensic.residual_private_v1 | veritrust-private/deepfake-forensic-residual-v1 @ UNKNOWN	private HF repo + HF Inference Endpoint custom container	requires_research_and_training
deepfake OOD	deepfake.ood.siglip2_private_v1 | veritrust-private/deepfake-ood-siglip2 @ UNKNOWN	private HF repo + HF Inference Endpoint	requires_training
full-frame same-family synthesis shadow	deepfake.synthetic.commfor224_shadow_v1 | OwensLab/commfor-model-224 @ 26afc31e6b40c312c3fd42c05a758be62446215b	HF Inference Endpoint	shadow_only
full-frame synthesis	deepfake.synthetic.commfor384_v1 | OwensLab/commfor-model-384 @ 6076002bf0d9dd37537f965ee2f06f826c333b61	HF Inference Endpoint	research_candidate
optional learning recommendation	learning.recommendation.private_v1 | veritrust-private/learning-recommendation-v1 @ UNKNOWN	private HF repo + HF Inference Endpoint	optional_future
link screenshot brand embedding	link.brand.siglip2_v1 | google/siglip2-base-patch16-224 @ 75de2d55ec2d0b4efc50b3e9ad70dba96a7b2fa2	HF Inference Endpoint	candidate_pending_benchmark

Role	Canonical model key / repository / revision	Deployment	Status
link structured features	link.features.gbdt_private_v1 | veritrust-private/link-feature-gbdt-v1 @ UNKNOWN	private HF repo + HF Inference Endpoint custom container	requires_training
URL lexical shadow	link.lexical.pirocheto_onnx_shadow_v1 | pirocheto/phishing-url-detection @ 44f3b19f705b52532e0aadf3d0d15dd892b8a2fb	HF Inference Endpoint custom container	research_only
malicious page text	link.page_text.deberta_private_v1 | veritrust-private/malicious-page-deberta-v1 @ UNKNOWN	private HF repo + HF Inference Endpoint	requires_training
malicious URL primary	link.url.urlbert_v4_v1 | CrabInHoney/urlbert-tiny-v4-malicious-url-classifier @ 917ded0543a630d6e82570bb01ae692f6cbb95f1	HF Inference Endpoint	candidate_pending_benchmark
phishing structured features	phishing.features.gbdt_private_v1 | veritrust-private/phishing-feature-gbdt-v1 @ UNKNOWN	private HF repo + HF Inference Endpoint custom container	requires_training
mail text shadow	phishing.text.cybersectony_shadow_v1 | cybersectony/phishing-email-detection-distilbert_v2.4.1 @ 6724dec17360ab1f6450553f1d950404f66a5dde	HF serverless shadow	shadow_only
phishing text fast candidate	phishing.text.ealvaradob_bert_v1 | ealvaradob/bert-finetuned-phishing @ fa8fb73a007174c410ab7160d4e4c6e6b8d998d4	HF serverless	candidate_pending_benchmark
phishing text fast candidate	phishing.text.fast.phishlens_v1 | Sonje03/phishlens-distilbert @ b585a2b9bbf6777b344d017f4eaff9423363a489	HF Inference Endpoint	candidate_promotion_gated
multilingual phishing	phishing.text.mdeberta_private_v1 | veritrust-private/phishing-mdeberta-v1 @ UNKNOWN	private HF repo + HF Inference Endpoint	requires_training
phishing semantic shadow	phishing.text.modernbert_shadow_v1 | mikaelnurminen/modernbert-large-phishing @ d28c16f882454115d71375d3321fbfd90e566490	HF Inference Endpoint	shadow_only

5. Model registry and pre-dispatch resolution

Current gateway inference occurs before gateway_model_versions lookup. That provenance race is a P0/P1 blocker.

A registry entry expresses scientific identity; a deployment expresses one executable HF target. Tenant policy, requested role, eligibility, active deployment, exact revision and schema/calibration versions are resolved and persisted before network I/O. Mutable display aliases never authorize execution.

Registry entry	Deployment
role, repo, revision, task, schemas, preprocessing, labels, calibration, benchmark, license/privacy	provider/endpoint, endpoint revision, region, instance, replicas, scale-to-zero, health
stable scientific identity	rotatable operational identity

6. Evidence contracts and inference receipts

Raw output, normalized semantics and calibrated probability remain separate nullable fields. Missingness is explicit. Completed learned evidence requires a matching pre-dispatch resolution and inference_receipt.v2. Every learned receipt carries a deployment_bundle_id and an ordered non-empty components[] list; the legacy v1 adapter never invents missing provenance.

Evidence family	Fields
Required fields	evidence_id, investigation_id, artifact_id, specialist, scientific_role, registry_resolution_id, deployment_id, model_key, immutable_model_revision, preprocessing_version, input_hash, raw_output, raw_label, raw_score, normalized_label, normalized_score, calibrated_probability, calibration_version, quality, sufficiency, ood_status, status, reason_codes, started_at, completed_at
Lifecycle states (16)	planned, dispatched, running, completed, abstained, not_applicable, unavailable, timed_out, failed, cancelled, privacy_blocked, policy_skipped, insufficient, unsupported, degraded, budget_exceeded
Semantic rule	raw_score, normalized_score, and calibrated_probability are separate nullable fields and never silently substituted.

Receipt family	Fields
inference_receipt.v2 fields	receipt_id, registry_resolution_id, deployment_id, deployment_bundle_id, request_id, provider, endpoint_identity, endpoint_deployment_revision, request_sha256, response_sha256, provider_request_id, components, attempt, latency_ms, billed_units, completion_status, received_at
components[] fields	component_role, model_key, repository_id, immutable_revision, artifact_or_container_digest, preprocessing_version, input_schema_version, output_schema_version, execution_order, qualification_status
Compatibility	A valid v1 receipt is exposed as a v2 receipt with one synthetic component derived from its singular repository/model/revision/preprocessing fields; missing historical provenance remains explicit and is never fabricated.

7. Fast, hybrid and deep paths

Fast text/URL work may complete inside a 12-second caller budget. Hybrid work returns 202 when it cannot complete. Deep media work is asynchronous and must not occupy a Vercel function for model execution.

Mode	Use	Behavior
Synchronous fast	qualified phishing text, URL primary	bounded request; completed evidence/policy only
Hybrid	multilingual/page/quality	accepted scan; continue queued
Asynchronous deep	deepfake roles	signed upload, poll/webhook, per-stage terminal state

8. Gateway, correlation, policy and review

Correlation consumes validated evidence envelopes; it does not average incomparable scores. Relationship rules, sufficiency and policy are independently versioned. Manual review is mandatory for high-consequence incomplete evidence, deepfake identity/financial contexts, strong disagreement, OOD, provider degradation, unsupported artifacts, low calibration confidence and critical actions.

Decision	Meaning
allow	policy permits; sufficient qualified evidence
review/hold	human action or additional evidence required
block	policy-authorized prevention with required evidence
uncertain/unsupported/failed	no fabricated safe or malicious conclusion

9. Database, workers and clients

The repository has a schema manifest, not live executable migrations. SQL must be derived and reviewed against the canonical database.

The target adds registry entries/deployments/resolutions, receipts, calibration artifacts, benchmark runs, provider incidents, promotions and human reviews. Workers orchestrate queues and remote HF calls but never load weights. Current Gateway routes and browser CLI/PowerShell flows remain compatible; evidence/review/provider-health routes are explicitly proposed.

10. Security, privacy and supply chain

Endpoint tokens are server-only secret references. Payloads are minimized per scientific role; routine logs exclude message bodies, media, URL secrets and tokens. Third-party providers require retention, training-use, residency, subprocessor and DPA review. Custom handlers/containers require code review, dependency locks, SBOM, scans and image digest.

- Protected endpoint by default; private connectivity when policy requires and HF supports it.
- No automatic third-party provider failover.
- Historical evidence and receipts are append-only.

11. Market / competitor lessons to architecture response

VeriTrust is differentiated by registry-bound scientific roles, explicit missingness, evidence receipts and Gateway-owned policy across Link, Phishing, Deepfake and Learning. These are responses to public market patterns, not claims about proprietary competitor internals.

Market pattern	VeriTrust response
Multi-signal email security	typed evidence graph, not one text model
URL reputation/intelligence	freshness-aware connector lane
Deepfake detection APIs	asynchronous receipt-bound evidence plus uncertainty
C2PA/content provenance	separate provenance evidence lane

Market pattern	VeriTrust response
Security-awareness platforms	consented advisory learning integration

12. Scaling, cost and observability

Use per-tenant rate/concurrency limits, weighted fair queues, backpressure, bounded retries, circuit breakers and artifact/contract deduplication. Scale-to-zero only where cold starts meet the role SLO. Attribute endpoint minutes or provider units to tenant/model/deployment. Tier numbers below are test envelopes, not measured capacity or SLOs.

Observe	Examples
Operational	rate, p50/p95/p99 latency, queue age, deadline miss, 429/5xx/cold starts
Contract	schema/label/receipt mismatch, unsupported inputs
Scientific	sufficiency, abstention, OOD, drift, calibration
Financial	per-request/provider/tenant cost and budget alerts

Tier	Planning load	HF topology	Cost/shadow gate
T0 research	1 rpm / 1 concurrent (planning cap)	serverless where qualified; endpoints scale to zero	100% shadow permitted; cost UNKNOWN until test
T1 pilot	10 rpm / 3 concurrent (planning cap)	fast serverless; protected endpoints 0-1 min replica by role	25% shadow; measured invoice required
T2 small production	60 rpm / 10 concurrent (planning cap)	qualified fast + bounded endpoint pools	10% shadow; tenant/model budgets
T3 medium	600 rpm / 50 concurrent (planning envelope)	load-derived replica ranges and weighted queues	2-5% shadow; reserved budget/alerts
T4 enterprise	>600 rpm / 100+ concurrent (design input)	account/region/SLA-specific pools and isolation	<=1% shadow; contract and load test required

13. Rollout, rollback and first tickets

P0 fixes semantics/fallbacks. P1 builds registry/contracts/receipts. P2 qualifies HF deployments. P3 benchmarks/calibrates/shadows. P4 tenant canary. P5 controlled production. P6 completes migration/provenance/learning extensions.

- First milestone: VT2-001 through VT2-016.
- Atomic registry pointer rollback; in-flight requests stay bound to their persisted deployment.
- No promotion without contract, benchmark, calibration, robustness, privacy, security, license, load, cost, observability and rollback gates.

14. Definition of done

Done requires no learned inference or weights in VeriTrust; pre-dispatch resolution plus bundle/component receipts; revision-bound labels and explicit missingness; compatible API/CLI/PowerShell clients; and live-schema-reviewed database changes. Runbooks, budgets, incident drills and rollback evidence remain promotion gates.

References and package evidence

Primary sources and immutable model snapshots are listed in Research/Sources/source_ledger.csv and Research/Sources/HuggingFace/. Repository claims are traceable through repository_inventory.csv, source_file_audit.csv, runtime_call_graph.md, current_architecture_map.md and code_claim_ledger.csv.

- Architecture_Source_of_Truth.yaml - canonical invariants, roles, schemas and rollout.
- Model_Registry_Proposal.json - immutable role/deployment proposal.
- HuggingFace_Deployment_Matrix.csv - point-in-time provider and endpoint decisions.
- Technical_Findings_Register.csv / Implementation_Backlog.csv - evidence-backed remediation.
- Database_Target_Contract.md / Database_Migration_Specification.md - logical contract without fabricated SQL.
- Competitor_Analysis.md / Competitor_Matrix.csv - primary-source market design loop.

Authoritative external references

Hugging Face Inference Providers | Hugging Face Inference Endpoints | C2PA Technical Specification. Access dates, exact page URLs, claim summaries and verification status are preserved in the source ledger.

Universal operational controls

Security, privacy, cost, capacity, calibration, observability, incident response and rollback are release gates for every role. No UNKNOWN is replaced with an estimate; unresolved values remain explicit until an evidence-producing test or owner decision closes them.

Final boundary

Hugging Face-managed learned inference is authoritative. No learned production model runs inside the VeriTrust Vercel application or a VeriTrust-owned worker/server; no production weights are required in the VeriTrust deployment repository.